

```
In [1]: import numpy as np
import pandas as pd

df = pd.read_csv(r"C:\Users\Administrator\Downloads\anivin.csv.txt")
df
```

```
Out[1]:
```

	id	name	department	age	salary
0	1	John	IT	25.0	50000.0
1	2	NaN	HR	30.0	60000.0
2	3	Mary	Finance	NaN	55000.0
3	4	Tom	IT	28.0	NaN
4	5	Alice	HR	32.0	62000.0
5	6	Bob	NaN	29.0	58000.0
6	7	John	IT	25.0	50000.0
7	8	gokul	Finance	31.0	61000.0

```
In [3]: df.isna().sum()
```

```
Out[3]: id          0
name          1
department    1
age           1
salary        1
dtype: int64
```

```
In [5]: data = df.dropna()
```

```
In [6]: data
```

```
Out[6]:
```

	id	name	department	age	salary
0	1	John	IT	25.0	50000.0
4	5	Alice	HR	32.0	62000.0
6	7	John	IT	25.0	50000.0
7	8	gokul	Finance	31.0	61000.0

```
In [11]: data['tax']=data['salary']*1.18
data
```

C:\Users\Administrator\AppData\Local\Temp\ipykernel_14952\2873981719.py:1: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
data['tax']=data['salary']*1.18
```

```
Out[11]:
```

	id	name	department	age	salary	tax
0	1	John	IT	25.0	50000.0	59000.0
4	5	Alice	HR	32.0	62000.0	73160.0
6	7	John	IT	25.0	50000.0	59000.0
7	8	gokul	Finance	31.0	61000.0	71980.0

```
In [10]: import sqlite3
conn = sqlite3.connect("sales.db")
data.to_sql("sales_table",conn,if_exists="replace",index=False)
print("Data loaded successfully")
```

Data loaded successfully

In []: Practice Questions:

- What is ETL ?

ETL stands for Extract, Transform, and Load. It is a process used in data integration. Data is extracted from different sources, transformed into a suitable format, and loaded into a database or data warehouse for analysis.

- What are the three stages of ETL?

Extract

Data is collected from various sources such as CSV files, databases, or APIs.

Transform

The extracted data is cleaned and modified. Examples include removing null values, calculating new fields, and formatting data.

Load

The transformed data is stored into a target system such as a SQL database or data warehouse.

- What is data transformation?

Data transformation is the process of converting raw data into a meaningful format. It may include filtering data, removing null values, calculating new columns, and changing data types.

- Difference between full load and incremental load?

Full Load Incremental Load

Loads the entire dataset every time

Loads only new or changed data

Takes more time Faster and efficient

Used for small datasets Used for large datasets

- Name two ETL Tools?

1. Microsoft SQL Server Integration Services (SSIS)
2. Informatica PowerCenter

Other examples: Talend, Apache NiFi